



FACULTY OF AGRICULTURE

BHABHA UNIVERSITY

Jatkhedi, NH-12 , Hoshangabad Road , Bhopal,
Madhya Pradesh- 462026

Bachelor of Science in Agriculture

SEMESTER-WISE DISTRIBUTION OF COURSEs

SCHEME

SEMESTER-V			
S.NO.	SUBJECT TITLE	SUBJECT CODE	CREDITS
1.	Principles of Integrated Pest and Disease Management	BAG-5501	3 (2+1)
2.	Manures, Fertilizers and Soil Fertility Management	BAG-5502	3 (2+1)
3.	Pests of Crops and Stored Grain and their Management	BAG-5503	3 (2+1)
4.	Diseases of Field and Horticultural Crops and their Management -I	BAG-5504	3 (2+1)
5.	Crop Improvement-I (Kharif Crops)	BAG-5505	2 (1+1)
6.	Entrepreneurship Development and Business Communication	BAG-5506	2 (1+1)
7.	Geoinformatics and Nano-technology and Precision Farming	BAG-5507	2 (1+1)
8.	Practical Crop Production – I (Kharif crops)	BAG-5508	2 (0+2)
9.	Intellectual Property Rights	BAG-5509	1 (1+0)
10.	Elective Course	BAG-5510	3 credit
TOTAL			21(12+09)+3



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SEMESTER-VI			
S.NO.	SUBJECT TITLE	SUBJECT CODE	CREDITS
1.	Rainfed Agriculture & Watershed Management	BAG-5601	2 (1+1)
2.	Protected Cultivation and Secondary Agriculture	BAG-5602	2 (1+1)
3.	Diseases of Field and Horticultural Crops and their Management-II	BAG-5603	3 (2+1)
4.	Post-harvest Management and Value Addition of Fruits and Vegetables	BAG-5604	2 (1+1)
5.	Management of Beneficial Insects	BAG-5605	2 (1+1)
6.	Crop Improvement-II (Rabi crops)	BAG-5606	2 (1+1)
7.	Practical Crop Production –II (Rabi crops)	BAG-5607	2 (0+2)
8.	Principles of Organic Farming	BAG-5608	2 (1+1)
9.	Farm Management, Production & Resource Economics	BAG-5609	2 (1+1)
10.	Principles of Food Science and Nutrition	BAG-5610	2 (2+0)
11.	Elective Course	BAG-5611	3 credit
TOTAL			21 (11+10)+3



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THIRD YEAR

SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Principles of Integrated Pest and Disease Management	BAG-5501	3(2+1)

THEORY

UNIT I

Categories of insect pests and diseases, IPM: Introduction, history, importance, concepts, principles and tools of IPM.

UNIT II

Economic importance of insect pests, diseases and pest risk analysis. Methods of detection and diagnosis of insect pest and diseases. Calculation and dynamics of economic injury level and importance of Economic threshold level.

UNIT III

Methods of control: Host plant resistance, cultural, mechanical, physical, legislative, biological and chemical control. Ecological management of crop environment. Introduction to conventional pesticides for the insect pests and disease management.

UNIT IV

Survey surveillance and forecasting of Insect pest and diseases. Development and validation of IPM module.

UNIT V

Implementation and impact of IPM (IPM module for Insect pest and disease. Safety issues in pesticide uses. Political, social and legal implication of IPM. Case histories of important IPM programmes.



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PRACTICAL

Methods of diagnosis and detection of various insect pests, and plant diseases, Methods of insect pests and plant disease measurement, Assessment of crop yield losses, calculations based on economics of IPM, Identification of biocontrol agents, different predators and natural enemies. Mass multiplication of *Trichoderma*, *Pseudomonas*, *Trichogramma*, NPV etc. Identification and nature of

damage of important insect pests and diseases and their management. Crop (agroecosystem) dynamics of a selected insect pest and diseases. Plan & assess preventive strategies (IPM module) and decision making. crop monitoring attacked by insect, pest and diseases . Awareness campaign at farmers' fields.



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SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Manures, Fertilizers and Soil Fertility Management	BAG-5502	3(2+1)

THEORY

UNIT I

Introduction and importance of organic manures, properties and methods of preparation of bulky and concentrated manures. Green/leaf manuring. Fertilizer recommendation approaches. Integrated nutrient management.

UNIT II

Chemical fertilizers: classification, composition and properties of major nitrogenous, phosphatic, potassic fertilizers, secondary & micronutrient fertilizers, Complex fertilizers, nano fertilizers Soil amendments, Fertilizer Storage, Fertilizer Control Order.

UNIT III

History of soil fertility and plant nutrition. criteria of essentiality. role, deficiency and toxicity symptoms of essential plant nutrients, Mechanisms of nutrient transport to plants, factors affecting nutrient availability to plants.

UNIT IV

Chemistry of soil nitrogen, phosphorus, potassium, calcium, magnesium, sulphur and micronutrients. Soil fertility evaluation, Soil testing. Critical levels of different nutrients in soil.

UNIT V

Forms of nutrients in soil, plant analysis, rapid plant tissue tests. Indicator plants. Methods of fertilizer recommendations to crops. Factor influencing nutrient use efficiency (NUE), methods of application under rainfed and irrigated conditions.

PRACTICAL

- Introduction of analytical instruments and their principles, calibration and applications
- Colorimetry and flame photometry
- Estimation of soil organic carbon
- Estimation of alkaline hydrolysable N in soils



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- Estimation of soil extractable P in soils
- Estimation of exchangeable K; Ca and Mg in soils
- Estimation of soil extractable S in soils
- Estimation of DTPA extractable Zn in soils
- Estimation of N in plants
- Estimation of P in plants
- Estimation of K in plants
- Estimation of S in plants

SYLLABUS

THIRD YEAR

SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Pests of Crops and Stored Grains and their Management	BAG-5503	3(2+1)

THEORY

UNIT I

General account on nature and type of damage by different arthropods pests. Scientific name, order, family, host range, distribution, biology and bionomics, nature of damage.

UNIT II

Management of major pests and scientific name, order, family, host range, distribution, nature of damage and control practice other important arthropod pests of various field crop, vegetable crop.

UNIT III

Management of major pests and scientific name, order, family, host range, distribution, nature of damage and control practice other important arthropod pests of various fruit crop, plantation crops, ornamental crops, spices and condiments.

UNIT IV

Factors affecting losses of stored grain and role of physical, biological, mechanical and chemical factors in deterioration of grain.

UNIT V

Insect pests, mites, rodents, birds and microorganisms associated with stored grain and their management. Storage structure and methods of grain storage and fundamental principles of grain



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store management.

PRACTICAL

Identification of different types of damage. Identification and study of life cycle and seasonal history of various insect pests attacking crops and their produce: (a) Field Crops; (b) Vegetable Crops; (c) Fruit Crops; (d) Plantation, gardens, Narcotics, spices & condiments. Identification of insect pests and Mites associated with stored grain. Determination of insect infestation by different methods. Assessment of losses due to insects. Calculations on the doses of insecticides application technique. Fumigation of grain store / godown. Identification of rodents and rodent control operations in godowns. Identification of birds and bird control operations in godowns. Determination of moisture content of grain. Methods of grain sampling under storage condition. Visit to Indian Storage Management and Research Institute, Hapur and Quality Laboratory, Department of Food., Delhi. Visit to nearest FCI godowns.

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THIRD YEAR

SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Diseases of Field and Horticultural Crops and their Management -I	BAG-5504	3(2+1)

THEORY

Unit I

Horticultural Crops: Guava: wilt, anthracnose; Banana: Panama wilt, bacterial wilt, Sigatoka and bunchy top, Papaya: foot rot, leaf curl and mosaic, Pomegranate: bacterial blight; Cruciferous vegetables: Alternaria leaf spot and black rot; Brinjal: Phomopsis blight and fruit rot and Sclerotinia blight; Tomato: damping off, wilt, early and late blight, buck eye rot and leaf curl and mosaic; Okra: Yellow Vein Mosaic

Unit II

Beans: anthracnose and bacterial blight; Ginger: soft rot; Colocasia: Phytophthora blight; Coconut: wilt and bud rot; Tea: blister blight; Coffee: rust

Unit III

Symptoms, etiology, disease cycle and management of major diseases of following crops:
Field



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Crops: Rice: blast, brown spot, bacterial blight, sheath blight, false smut, khaira and tungro; Maize: stalk rots, downy mildew, leaf spots; Sorghum: smuts, grain mold and anthracnose, Bajra :downy mildew and ergot; Groundnut: early and late leaf spots, wilt

Unit IV

Soybean: Rhizoctonia blight, bacterial spot, seed and seedling rot and Pigeonpea: Phytophthora blight, wilt and sterility mosaic;

Unit V

Finger millet: Blast and leaf spot; black & green gram: Cercospora leaf spot and anthracnose, web blight and yellow mosaic; Castor: Phytophthora blight; Tobacco: black shank, black root rot and mosaic.

PRACTICAL

Identification and histopathological studies of selected diseases of field and horticultural crops covered in theory. Field visit for the diagnosis of field problems. Collection and preservation of plant diseased specimens for Herbarium; Note: Students should submit 50 pressed and well-mounted specimens.

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SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Crop Improvement-I (Kharif Crops)	BAG-5505	2(1+1)

THEORY

Unit I

Plant genetic resources; its utilization and conservation. Study of genetics of qualitative and quantitative characters.

Unit II

Important concepts of breeding self pollinated, cross pollinated and vegetatively propagated crops. Floral biology, emasculation, pollination, centers of origin, distribution of species, wild relatives of different cereals and pulses of *kharif* season.

Unit III



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Floral biology, emasculation, pollination, centers of origin, distribution of species, wild relatives of different oilseeds, fibres, fodders and cash crops, vegetable and horticultural crops of *kharif* season.

Unit IV

Major breeding objectives and procedures including conventional and modern innovative approaches for development of hybrids and varieties for yield, adaptability, stability, abiotic and biotic stress tolerance and quality (physical, chemical, nutritional) of different *kharif* crops.

Unit V

Hybrid seed production technology in rice, maize, sorghum, pearl millet and pigeon pea. Ideotype concept and climate resilient crop varieties for future.

PRACTICAL

- Floral biology, emasculation and hybridization techniques in different crop species viz., Rice, Jute, Maize, Sorghum, Pearl millet, Ragi, Pigeonpea, Urdbean, Mungbean, Soybean, Groundnut, Sesame, Caster, Cotton, Cowpea, Tobacco, Brinjal, Okra and Cucurbitaceous crops.
- Maintenance breeding of different *kharif* crops.
- Handling of germplasm and segregating populations by different methods like pedigree, bulk and single seed decent methods.
- Study of field techniques for seed production and hybrid seeds production in *Kharif* crops.
- Estimation of heterosis, inbreeding depression and heritability.
- Layout of field experiments.
- Study of quality characters.
- Donor parents for different characters.
- Visit to seed production plots.
- Visit to AICRP plots of different field crops.

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SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Entrepreneurship Development and Business Communication	BAG-5506	2(1+1)

THEORY

Unit I Concept of Entrepreneur, Entrepreneurship Development, Characteristics of entrepreneurs;



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SWOT Analysis & achievement motivation, Government policy and programs and institutions for entrepreneurship development, Impact of economic reforms on Agribusiness/Agrienterprises,

Unit III

Entrepreneurial Development Process; Business Leadership Skills; Developing organizational skill (controlling, supervising, problem solving, monitoring & evaluation),

Unit IV

Developing Managerial skills, Business Leadership Skills (Communication, direction and motivation Skills), Problem solving skill, Supply chain management and Total quality management,

Unit V

Project Planning Formulation and report preparation; Financing of enterprise, Opportunities for agripreneurship and rural enterprise.

PRACTICAL

Assessing entrepreneurial traits, problem solving skills, managerial skills and achievement motivation, exercise in creativity, time audit through planning, monitoring and supervision, identification and selection of business idea, preparation of business plan and proposal writing, visit to entrepreneurship development institute and entrepreneurs.

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THIRD YEAR

SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Geoinformatics and Nano-technology and Precision Farming	BAG-5507	2(1+1)

THEORY

Unit I

Precision agriculture: concepts and techniques; their issues and concerns for Indian agriculture, Geo- informatics- definition, concepts, tool and techniques; their use in Precision Agriculture.

Unit II

Crop discrimination and Yield monitoring, soil mapping; fertilizer recommendation using geospatial technologies; Spatial data and their management in GIS;



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Unit III

Remote sensing concepts and application in agriculture; Image processing and interpretation; Global positioning system (GPS), components and its functions;

Unit IV

Introduction to crop Simulation Models and their uses for optimization of Agricultural Inputs; STCR approach for precision agriculture;

Unit V

Nanotechnology, definition, concepts and techniques, brief introduction about nanoscale effects, nano-particles, nano-pesticides, nano-fertilizers, nano-sensors, Use of nanotechnology in seed, water, fertilizer, plant protection for scaling-up farm productivity.

PRACTICAL

Introduction to GIS software, spatial data creation and editing. Introduction to image processing software. Visual and digital interpretation of remote sensing images. Generation of spectral profiles of different objects. Supervised and unsupervised classification and acreage estimation. Multispectral remote sensing for soil mapping. Creation of thematic layers of soil fertility based on GIS. Creation of productivity and management zones. Fertilizers recommendations based of VRT and STCR techniques. Crop stress (biotic/abiotic) monitoring using geospatial technology. Use of GPS for agricultural survey. Formulation, characterization and applications of nanoparticles in agriculture. Projects formulation and execution related to precision farming.

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THIRD YEAR		SEMESTER V
SUBJECT TITLE	SUBJECT CODE	CREDIT
Practical Crop Production-I (<i>Kharif Crops</i>)	BAG-5508	2(0+2)

PRACTICAL

Unit I

Crop planning, raising field crops in multiple cropping systems: Field preparation, seed, treatment, nursery raising, sowing, nutrient, water and weed management and management of insect-pests diseases of crops, harvesting, threshing, drying winnowing, storage and marketing of produce.

Unit II



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The emphasis will be given to seed production, mechanization, resource conservation and integrated nutrient, insect-pest and disease management technologies.

Unit III

Preparation of balance sheet including cost of cultivation, net returns per student as well as per team of 8-10 students.

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THIRD YEAR

SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Intellectual Property Rights	BAG-5509	1(1+0)

THEORY

UNIT I

Introduction and meaning of intellectual property, brief introduction to GATT, WTO, TRIPs and WIPO, Treaties for IPR protection: Madrid protocol, Berne Convention, Budapest treaty, etc.

UNIT II

Types of Intellectual Property and legislations covering IPR in India:-Patents, Copyrights, Trademark, Industrial design, Geographical indications, Integrated circuits, Trade secrets.

UNIT III

Patents Act 1970 and Patent system in India, patentability, process and product patent, filing of patent, patent specification, patent claims, Patent opposition and revocation, infringement, Compulsory licensing, Patent Cooperation Treaty, Patent search and patent database.

UNIT IV

Origin and history including a brief introduction to UPOV for protection of plant varieties, Protection of plant varieties under UPOV and PPV&FR Act of India, Plant breeders rights, Registration of plant varieties under PPV&FR Act 2001, breeders, researcher and farmers rights.

UNIT V



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Traditional knowledge-meaning and rights of TK holders. Convention on Biological Diversity, International treaty on plant genetic resources for food and agriculture (ITPGRFA). Indian Biological Diversity Act, 2002 and its salient features, access and benefit sharing.

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SEMESTER V

SUBJECT TITLE	SUBJECT CODE	CREDIT
Agrochemicals(Elective Course)	BAG-5510	3(2+1)

THEORY

UNIT I

An introduction to agrochemicals, their type and role in agriculture, effect on environment, soil, human and animal health, merits and demerits of their uses in agriculture, management of agrochemicals for sustainable agriculture.

UNIT II

Herbicides-Major classes, properties and important herbicides. Fate of herbicides. Fertilizers and their importance. Nitrogenous fertilizers: Feedstocks and Manufacturing of ammonium sulphate, ammonium nitrate, ammonium chloride, urea. Slow release N-fertilizers.

UNIT III

Fungicides- Classification – Inorganic fungicides - characteristics, preparation and use of sulfur and copper, Mode of action-Bordeaux mixture and copper oxychloride. Organic fungicides- Mode of action- Dithiocarbamates-characteristics, preparation and use of Zineb and maneb. Systemic fungicides- Benomyl, carboxin, oxycarboxin, Metalaxyl, Carbendazim, characteristics and use.

UNIT IV

Introduction and classification of insecticides: inorganic and organic insecticides Organochlorine, Organophosphates, Carbamates, Synthetic pyrethroids Neonicotinoids, Biorationals, Insecticide Act and rules, Insecticides banned, withdrawn and restricted use, Fate of insecticides in soil & plant. IGRs Biopesticides, Reduced risk insecticides, Botanicals, plant and animal systemic insecticides their characteristics and uses. Plant bio-pesticides for ecological agriculture, Bio-insect repellent.

UNIT V



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Phosphatic fertilizers: feedstock and manufacturing of single superphosphate. Preparation of bone meal and basic slag. Potassic fertilizers: Natural sources of potash, manufacturing of potassium chloride, potassium sulphate and potassium nitrate. Mixed and complex fertilizers: Sources and compatibility—preparation of major, secondary and micronutrient mixtures. Complex fertilizers: Manufacturing of ammonium phosphates, nitrophosphates and NPK complexes. Fertilizer control order. Fertilizer logistics and marketing.

PRACTICAL

- Sampling of fertilizers and pesticides
- Pesticides application technology to study about various pesticides appliances
- Quick tests for identification of common fertilizers
- Identification of anion and cation in fertilizer
- Calculation of doses of insecticides to be used
- To study and identify various formulations of insecticide available in market
- Estimation of nitrogen in Urea
- Estimation of water soluble P_2O_5 and citrate soluble P_2O_5 in single super phosphate
- Estimation of potassium in Muriate of Potash/Sulphate of Potash by flame photometer
- Determination of copper content in copper oxychloride
- Determination of sulphur content in sulphur fungicide
- Determination of thiram
- Determination of ziram content



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SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Rainfed Agriculture and Watershed Management	BAG-5601	2 (1+1)

THEORY

Unit I

Rainfed agriculture: Introduction, types, History of rainfed agriculture and watershed in India; Problems and prospects of rainfed agriculture in India.

Unit II

Soil and climatic conditions prevalent in rainfed areas; Soil and water conservation techniques, Drought: types, effect of water deficit on physio-morphological characteristics of the plants

Unit III

Crop adaptation and mitigation to drought; Water harvesting: importance, its techniques

Unit IV

Efficient utilization of water through soil and crop management practices, Management of crops in rainfed areas.

Unit V

Contingent crop planning for aberrant weather conditions, Concept, objective, principles and components of watershed management, factors affecting watershed management.

PRACTICAL

Studies on climate classification, studies on rainfall pattern in rainfed areas of the country and pattern of onset and withdrawal of monsoons. Studies on cropping pattern of different rainfed areas in the country and demarcation of rainfed area on map of India. Interpretation of meteorological data and scheduling of supplemental irrigation on the basis of evapo-transpiration demand of crops. Critical analysis of rainfall and possible drought period in the country, effective rainfall and its calculation.



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Studies on cultural practices for mitigating moisture stress. Characterization and delineation of model watershed. Field demonstration on soil & moisture conservation measures. Field demonstration on construction of water harvesting structures. Visit to rainfed research station/watershed.



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THIRD YEAR

SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Protected Cultivation and Secondary Agriculture	BAG-5602	2 (1+1)

THEORY

Unit I

Green house technology: Introduction, Types of Green Houses; Plant response to Green house environment, Planning and design of greenhouses,

Unit II

Design criteria of green house for cooling and heating purposes. Green house equipments, materials of construction for traditional and low cost green houses.

Unit III

Irrigation systems used in greenhouses, typical applications, passive solar green house, hot air green house heating systems, green house drying. Cost estimation and economic analysis.

Unit IV

Important Engineering properties such as physical, thermal and aero & hydrodynamic properties of cereals, pulses and oilseed, their application in PHT equipment design and operation.

Unit V

Drying and dehydration; moisture measurement, EMC, drying theory, various drying method, commercial grain dryer (deep bed dryer, flat bed dryer, tray dryer, fluidized bed dryer, recirculatory dryer and solar dryer). Material handling equipment; conveyer and elevators, their principle, working and selection.



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PRACTICAL

Study of different type of green houses based on shape. Determine the rate of air exchange in an active summer winter cooling system. Determination of drying rate of agricultural products inside green house. Study of green house equipments. Visit to various Post Harvest Laboratories. Determination of Moisture content of various grains by oven drying & infrared moisture methods. Determination of engineering properties (shape and size, bulk density and porosity of biomaterials). Determination of Moisture content of various grains by moisture meter. Field visit to seed processing plant.



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SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Diseases of Field and Horticultural Crops and their Management-II	BAG-5603	3(2+1)

THEORY

Unit I

Symptoms, etiology, disease cycle and management of following diseases: Horticultural Crops:Mango: anthracnose, malformation, bacterial blight and powdery mildew; Citrus: canker and gummosis; Symptoms, etiology, disease cycle and management of following diseases: Horticultural Crops:Grape vine: downy mildew, Powdery mildew and anthracnose; Apple: scab, powdery mildew, fire blight and crown gall; Peach: leaf curl.

Unit II

Symptoms, etiology, disease cycle and management of following diseases: Horticultural Crops:Strawberry: leaf spot Potato: early and late blight, black scurf, leaf roll, and mosaic;

Unit III

Symptoms, etiology, disease cycle and management of following diseases: Horticultural Crops:Cucurbits: downy mildew, powdery mildew, wilt; Onion and garlic: purple blotch, and Stemphylium blight; Chilies: anthracnose and fruit rot, wilt and leaf curl; Turmeric: leaf spot Coriander: stem gall Marigold: Botrytis blight; Rose: dieback, powdery mildew and black leaf spot.

Unit IV

Symptoms, etiology, disease cycle and management of following diseases: Field Crops:Wheat: rusts, loose smut, karnal bunt, powdery mildew, alternaria blight, and ear cockle; Sugarcane: red rot, smut, wilt, grassy shoot, ratoon stunting and Pokkah Boeng;

Unit V

Symptoms, etiology, disease cycle and management of following diseases: Field Crops:Sunflower: Sclerotinia stem rot and Alternaria blight; Mustard: Alternaria blight, white rust, downy mildew and Sclerotinia stem rot; Gram: wilt, grey mould and Ascochyta blight; Lentil: rust and wilt; Cotton: anthracnose, vascular wilt, and black arm; Pea: downy mildew, powdery mildew and rust.

PRACTICAL

Identification and histopathological studies of selected diseases of field and horticultural crops covered in theory. Field visit for the diagnosis of field problems. Collection and preservation of plant diseased specimens for herbarium.



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SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Post-harvest Management and Value Addition of Fruits and Vegetables	BAG-5604	2(1+1)

THEORY

UNIT I

Importance of post-harvest processing of fruits and vegetables, extent and possible causes of post harvest losses; Pre-harvest factors affecting postharvest quality, maturity, ripening and changes occurring during ripening; Respiration and factors affecting respiration rate.

UNIT II

Harvesting and field handling; Storage (ZECC, cold storage, CA, MA, and hypobaric); Value addition concept; Principles and methods of preservation.

UNIT III

Intermediate moisture food- Jam, jelly, marmalade, preserve, candy – Concepts and Standards; Fermented and non-fermented beverages.

UNIT IV

Tomato products- Concepts and Standards; Drying/ Dehydration of fruits and vegetables – Concept and methods, osmotic drying.

UNIT V

Canning - Concepts and Standards, packaging of products

PRACTICAL

Applications of different types of packaging, containers for shelf life extension. Effect of temperature on shelf life and quality of produce. Demonstration of chilling and freezing injury in vegetables and fruits. Extraction and preservation of pulps and juices. Preparation of jam, jelly, RTS, nectar, squash, osmotically dried products, fruit bar and candy and tomato products, canned products. Quality evaluation of products -- physico-chemical and sensory. Visit to processing unit/ industry.



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SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Management of Beneficial Insects	BAG-5605	2(1+1)

THEORY

UNIT I

Importance of beneficial Insects, Beekeeping and pollinators, bee biology, commercial methods of rearing, equipment used, seasonal management, bee enemies and disease.

UNIT II

Bee pasturage, bee foraging and communication. Insect pests and diseases of honey bee. Role of pollinators in cross pollinated plants.

UNIT III

Types of silkworm, voltinism and biology of silkworm. Mulberry cultivation, mulberry varieties and methods of harvesting and preservation of leaves. Rearing, mounting and harvesting of cocoons.

Pest and diseases of silkworm, management, rearing appliances of mulberry silkworm and methods of disinfection.

UNIT IV

Species of lac insect, morphology, biology, host plant, lac production – seed lac, button lac, shellac, lac- products.

UNIT V

Identification of major parasitoids and predators commonly being used in biological control. Insect orders bearing predators and parasitoids used in pest control and their mass multiplication techniques. Important species of pollinator, weed killers and scavengers with their importance.

PRACTICAL

Honey bee species, castes of bees. Beekeeping appliances and seasonal management, bee enemies and disease. Bee pasturage, bee foraging and communication. Types of silkworm, voltinism and biology of silkworm. Mulberry cultivation, mulberry varieties and methods of harvesting and preservation of leaves. Species of lac insect, host plant identification. Identification of other important pollinators,



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weed killers and scavengers. Visit to research and training institutions devoted to beekeeping, sericulture, lac culture and natural enemies. Identification and techniques for mass multiplication of natural enemies.



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THIRD YEAR

SYLLABUS

SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Crop Improvement – II (<i>Rabi</i>)	BAG-5606	2(1+1)

THEORY

Unit I

Plant genetic resources, its utilization and conservation. Study of genetics of qualitative and quantitative characters.

Unit II

Floral biology, emasculation, pollination, centers of origin, distribution of species, wild relatives in different cereals and pulses of *Rabi* season.

Unit III

Floral biology, emasculation, pollination, centers of origin, distribution of species, wild relatives in different oilseeds, fodder crops and cash crops, vegetable and horticultural crops of *Rabi* season.

Unit IV

Major breeding objectives and procedures including conventional and modern innovative approaches for development of hybrids and varieties for yield, adaptability, stability, abiotic and biotic stress tolerance and quality (physical, chemical, nutritional) of different *Rabi* crops.

Unit V

Hybrid seed production technology of *rabi* crops. Ideotype concept and climate resilient crop varieties for future.

PRACTICAL

- Floral biology, emasculation and hybridization techniques in different crop species namely Wheat, Oat, Barley, Chickpea, Lentil, Field pea, Rajma, Horse gram, Rapeseed Mustard, Sunflower, Safflower, Potato, Berseem. Sugarcane, Tomato, Chilli, Onion.
- Handling of germplasm and segregating populations by different methods like pedigree, bulk and single seed decent methods.
- Study of field techniques for seed production and hybrid seeds production in *Rabi* crops.
- Estimation of heterosis, inbreeding depression and heritability.
- Layout of field experiments.
- Study of quality characters.



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- Study of donor parents for different characters.
- Visit to seed production plots.
- Visit to AICRP plots of different field crops.



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SYLLABUS

SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Practical Crop Production –II (Rabi crops)	BAG-5607	2(0+2)

PRACTICAL

Unit I

Crop planning, raising field crops in multiple cropping systems: Field preparation, seed, treatment, nursery raising, sowing, nutrient, water and weed management and management of insect-pests diseases of crops, harvesting, threshing, drying winnowing, storage and marketing of produce.

Unit II

The emphasis will be given to seed production, mechanization, resource conservation and integrated nutrient, insect-pest and disease management technologies.

Unit III

Preparation of balance sheet including cost of cultivation, net returns per student as well as per team of 8-10 students.



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SYLLABUS

SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Principles of Organic Farming	BAG-5608	2(1+1)

THEORY

Unit I

Organic farming, principles and its scope in India; Initiatives taken by Government (central/state), NGOs and other organizations for promotion of organic agriculture; Organic ecosystem and their concepts

Unit II

Organic nutrient resources and its fortification; Restrictions to nutrient use in organic farming.

Unit III

Choice of crops and varieties in organic farming; Fundamentals of insect, pest, disease and weed management under organic mode of production.

Unit IV

Operational structure of NPOP; Certification process and standards of organic farming.

Unit V

Processing, leveling, economic considerations and viability, marketing and export potential of organic products.

PRACTICAL

Visit of organic farms to study the various components and their utilization; Preparation of enrich compost, vermicompost, bio-fertilizers/bio-inoculants and their quality analysis; Indigenous technology knowledge (ITK) for nutrient, insect, pest disease and weed management; Cost of organic production system; Post harvest management; Quality aspect, grading, packaging and handling.



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SYLLABUS

SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Farm Management, Production & Resource Economics	BAG-5609	2(1+1)

THEORY

UNIT I

Meaning and concept of farm management, objectives and relationship with other sciences. Meaning and definition of farms, its types and characteristics, factor determining types and size of farms. Principles of farm management: concept of production function and its type, use of production function in decision-making on a farm, factor-product, factor-factor and product relationship, law of equi-marginal/or principles of opportunity cost and law of comparative advantage.

UNIT II

Meaning and concept of cost, types of costs and their interrelationship, importance of cost in managing farm business and estimation of gross farm income, net farm income, family labour income and farm business income.

UNIT III

Farm business analysis: meaning and concept of farm income and profitability, technical and economic efficiency measures in crop and livestock enterprises. Importance of farm records and accounts in managing a farm, various types of farm records needed to maintain on farm, farm inventory, balance sheet, profit and loss accounts.

UNIT IV

Meaning and importance of farm planning and budgeting, partial and complete budgeting, steps in farm planning and budgeting-linear programming, appraisal of farm resources, selection of crops and livestock's enterprises. Concept of risk and uncertainty occurs in agriculture production, nature and sources of risks and its management strategies, Crop/livestock/machinery insurance – weather based crop insurance, features, determinants of compensation.

UNIT V

Concepts of resource economics, differences between NRE and agricultural economics, unique properties of natural resources. Positive and negative externalities in agriculture, Inefficiency and



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welfare loss, solutions, Important issues in economics and management of common property resources of land, water, pasture and forest resources etc.

PRACTICAL

Preparation of farm layout. Determination of cost of fencing of a farm. Computation of depreciation cost of farm assets. Application of equi-marginal returns/opportunity cost principle in allocation of farm resources. Determination of most profitable level of inputs use in a farm production process. Determination of least cost combination of inputs. Selection of most profitable enterprise combination. Application of cost principles including CACP concepts in the estimation of cost of crop and livestock enterprises. Preparation of farm plan and budget, farm records and accounts and profit & loss accounts. Collection and analysis of data on various resources in India.



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SYLLABUS

SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Principles of Food Science and Nutrition	BAG-5610	2 (2+0)

THEORY

Unit I

Concepts of Food Science (definitions, measurements, density, phase change, pH, osmosis, surface tension, colloidal systems etc.)

Unit II

Food composition and chemistry (water, carbohydrates, proteins, fats, vitamins, minerals, flavours, colours, miscellaneous bioactives, important reactions);

Unit III

Food microbiology (bacteria, yeast, moulds, spoilage of fresh & processed foods, Production of fermented foods);

Unit IV

Principles and methods of food processing and preservation (use of heat, low temperature, chemicals, radiation, drying etc.);

Unit V

Food and nutrition, Malnutrition (over and undernutrition), nutritional disorders; Energy metabolism (carbohydrate, fat, proteins); Balanced/modified diets, Menu planning, New trends in food science and nutrition.



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SEMESTER VI

SUBJECT TITLE	SUBJECT CODE	CREDIT
Bio pesticides&Bio fertilizers (Elective Course)	BAG-5611	3(2+1)

THEORY

Unit I

History and concept of biopesticides.Importance, scope and potential of biopesticide.Definitions, concepts and classification of biopesticides viz. pathogen, botanical pesticides,

and biorationales.Botanicals and their uses.

Unit II

Biofertilizers - Introduction, status and scope. Structure and characteristic features of bacterial biofertilizers- *Azospirillum*, *Azotobacter*, *Bacillus*, *Pseudomonas*, *Rhizobium* and *Frankia*; Cynobacterial biofertilizers- *Anabaena*, *Nostoc*, *Hapalosiphon* and fungal biofertilizers- AM mycorrhiza and ectomycorrhiza.

Unit III

Mass production technology of bio-pesticides.Virulence, pathogenicity and symptoms of entomopathogenic pathogens and nematodes.Methods of application of biopesticides.Methods of quality control and Techniques of biopesticides.Impediments and limitation in production and use of biopesticide.

Unit IV

Nitrogen fixation -Free living and symbiotic nitrogen fixation.Mechanism of phosphate solubilization and phosphate mobilization, K solubilization. Production technology: Strain selection, sterilization, growth and fermentation, mass production of carrier based and liquid biofertilizers.

Unit V

FCO specifications and quality control of biofertilizers. Application technology for seeds, seedlings, tubers, sets etc. Biofertilizers -Storage, shelf life, quality control and marketing. Factors influencing the efficacy of biofertilizers.



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PRACTICAL

- Isolation and purification of important biopesticides: *Trichoderma Pseudomonas*, *Bacillus*, *Metarhyziium*etc. and its production
- Identification of important botanicals
- Visit to biopesticide laboratory in nearby area
- Field visit to explore naturally infected cadavers
- Identification of entomopathogenic entities in field condition
- Quality control of biopesticides
- Isolation and purification of *Azospirillum*, *Azotobacter*, *Rhizobium*, P-solubilizers and cyanobacteria
- Mass multiplication and inoculums production of biofertilizers
- Isolation of AM fungi -Wet sieving method and sucrose gradient method
- Mass production of AM inoculants